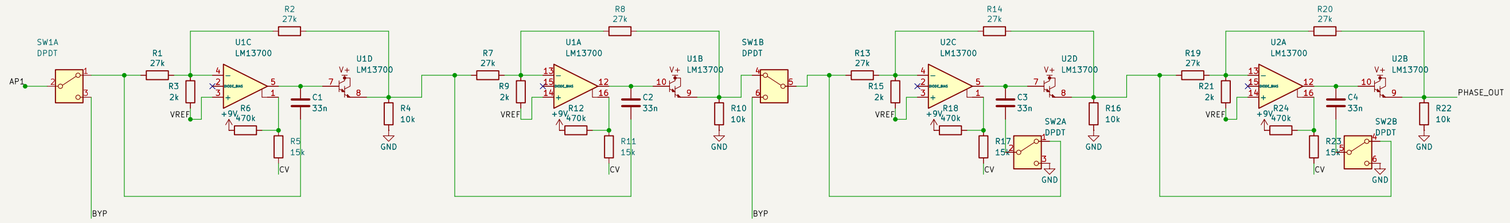
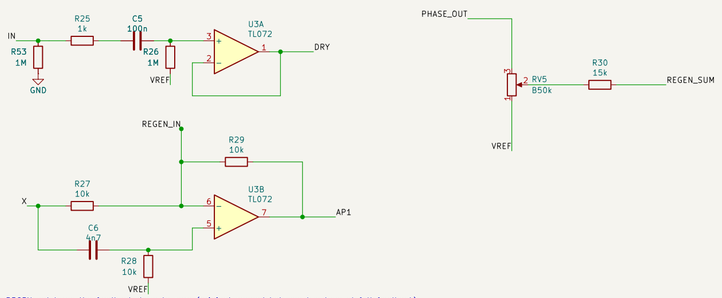
SWEPT ALL-PASS STAGES (LM13700 OTA + buffer, 27k/27k/2k, 33n)



STAGES: 6 = AP1 + S1..S4 + AP2, 4 = AP1 + S3 + S4 + AP2

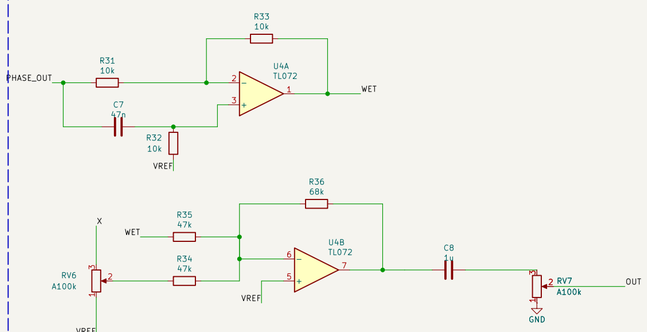
PHASE/FILTER: S3/S4 caps to the stage input (all-pass) or to ground (low-pass)

INPUT BUFFER, FIXED ALL-PASS 1 & REGEN (U3)



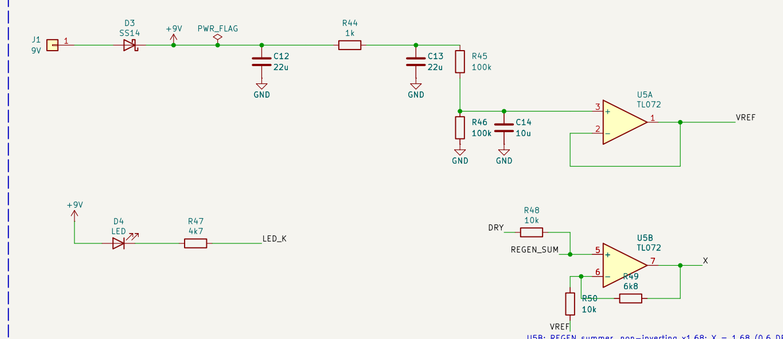
REGEN pot turns the feedback down to zero (original panned between two taps at full feedback)

FIXED ALL-PASS 2, MIX & LEVEL (U4)

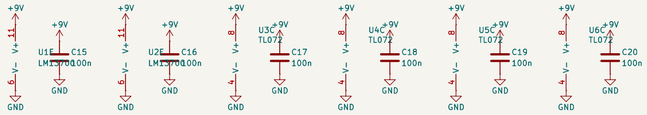


LEVEL = output volume (original: 1 + 68k/pot, unlimited gain at one end)

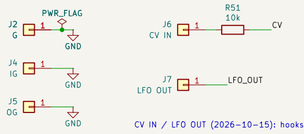
POWER, VREF & OFF-BOARD WIRING



U5B: REGEN summer, non-inverting x1.68: X = 1.68 (0.6 DRY + 0.4 REGEN). X



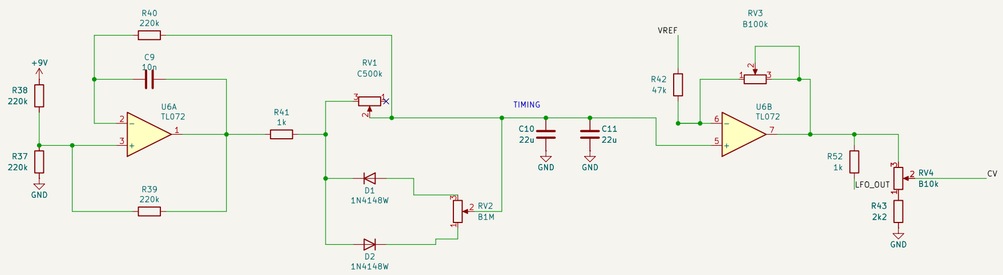




CV IN / LFO OUT (2026-10-15): hooks for a random / S&H LFO or othe

J1 = 9V, J2 = 9V ground, J4/J5 = jack grounds

LFO (U6A relaxation oscillator + U6B width amp) & CONTROL CURRENT



CV feeds each OTA through its own 10k: 470k from +9V keeps every OTA slightly on

Timing cap: 2x 22u ceramic (original 47u); ceramic derating makes the LFO a bit faster

TL072 LFO, one labc resistor per OTA + minimum current, leftovers removed
Fixes: MIX to VREF + inverting summer, LEVEL volume, REGEN level, buffered VREF, one polarity diode
After "Causality 6" by Rick Holt (2010) and the 2023 Eagle build

Super-Freq

Sheet: /
File: Casualty6.kicad_sch

Title: Casualty 6 SMD (Causality 6 phaser, fixed)

Size: A2	Date: 2026-09-26	**Rev: 1.0**
KiCad E.D.A. 10.0.4		Id: 1/1